“PAPER_{0:D3}” -f [int]# PAPER #78 □ Extragalactic Physics: NED Multi-Wavelength + UQFF

Title: NED Multi-Wavelength Extragalactic Physics: AGN Luminosity Functions and UQFF Buoyancy-Modified Hubble Tension Analysis

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Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57)
Date: March 7, 2026
Source Data: QCalc_validation.py (DataSourceURLs: NED_BASE, NED_API, QUASAR_SDSS)
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1. UQFF Hubble Constant Analysis

Hubble Tension Context

Measurement	H0 (km/s/Mpc)	Method
Planck 2018 (CMB)	67.4 ± 0.5	Early universe
SH0ES 2023 (Cepheids)	73.0 ± 1.0	Distance ladder
Tension	4.2s	□

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The UQFF vacuum buoyancy modifies the effective expansion rate:

$$H_{\text{UQFF}}(z) = H_0 \times \sqrt{\Omega_\Lambda + \Omega_m(1+z)^3 + [UA] \times \rho_{\text{vac,(UQFF)}} \times 8\pi G/3H_0^2}$$

The [UA] = 0.0001 fractional vacuum coupling adds:

$$\Delta H_0 = H_0 \times [UA] \times 0.5 = 67.4 \times 0.0001 \times 0.5 = 0.0034 \text{ km/s/Mpc}$$

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NED quasar catalog comparison:

Redshift bin	L*_standard (L?)	L*_UQFF (L?)	NED data range
z = 0.5	10^{45.0}	10^{45.3}	$10^{\{44.8\}□10}$ {45.5}
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UQFF correction to Hubble tension: ?H0 = 0.003 km/s/Mpc □ far too small to resolve the 5.6 km/s/Mpc tension. The UQFF does not attempt to resolve Hubble tension through the basic [UA] coupling; a higher-order Resonant Hubble correction would require additional development.

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2. AGN Luminosity Function Comparison

The UQFF Superconductive mode modifies the AGN accretion efficiency, shifting the break luminosity L*:

$$L_*^{\text{UQFF}} = L_*^{\text{standard}} \times (1 + [SCm]) = L_* \times 1.99$$

NED quasar catalog comparison:

Redshift bin	L*_standard (L?)	L*_UQFF (L?)	NED data range
z = 0.5	10^{45.0}	10^{45.3}	10^{{44.8}□10}{45.5}

Redshift bin	L*_standard (L?)	L*_UQFF (L?)	NED data range
z = 1.0	10^{45.5}	10^{45.8}	$10^{\{45.2\}□10}${46.0}
z = 2.0	10^{45.8}	10^{46.1}	$10^{\{45.5\}□10}${46.3}
z = 3.0	10^{46.0}	10^{46.3}	$10^{\{45.7\}□10}${46.5}

UQFF L* shift of 0.3 dex lies within the 0.5-dex observed scatter □ compatible with NED quasar data at all redshifts.

---

3. Quasar Absorption Line Systems

DLA (Damped Lyman-a) systems contain high column density neutral hydrogen (N_HI > $2\times10^{2°}$ cm?^{2}). The UQFF predicts no modification to the HI 21 cm line frequency (only gravitational Doppler at 10?$^{1°}$ level). NED DLA catalog: UQFF-consistent.

Summary

Observable	NED Data	UQFF Prediction	Agreement
H0 tension	5.6 km/s/Mpc gap	?H0 = 0.003 (negligible)	Not resolved
AGN L*	$10^{\{45\}□10}${46.5}	+0.3 dex ([SCm])	Within scatter
DLA HI column	N_HI > 10^{20}	Unmodified	Compatible

Source: QCalc_validation.py NED_BASE endpoint | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER_{0:D3}" -f $n

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Abstract

The NED (NASA/IPAC Extragalactic Database) multi-wavelength catalog covers UV through radio for >1 billion extragalactic objects. Key physics tests for UQFF: (1) AGN luminosity functions comparing UQFF-enhanced accretion vs standard models, (2) the Hubble tension (H0 = 67□73 km/s/Mpc) examined through the UQFF Buoyant vacuum correction, and (3) quasar absorption line systems (DLA/LLS) testing the UQFF vacuum density at cosmological redshifts.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?^{1}, [SSq] = 0.57) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. UQFF Hubble Constant Analysis

Hubble Tension Context

Measurement	H0 (km/s/Mpc)	Method
Planck 2018 (CMB)	67.4 ± 0.5	Early universe
SH0ES 2023 (Cepheids)	73.0 ± 1.0	Distance ladder
Tension	4.2s	□

UQFF Buoyant Correction to H0

The UQFF vacuum buoyancy modifies the effective expansion rate:

$$H_{\text{UQFF}}(z) = H_0 \times \sqrt{\Omega_\Lambda + \Omega_m(1+z)^3 + [UA] \times \rho_{\text{vac,(UQFF)}} \times 8\pi G/3H_0^2}$$

The [UA] = 0.0001 fractional vacuum coupling adds:

$$\Delta H_0 = H_0 \times [UA] \times 0.5 = 67.4 \times 0.0001 \times 0.5 = 0.0034 \text{ km/s/Mpc}$$

UQFF correction to Hubble tension: ?H0 = 0.003 km/s/Mpc □ far too small to resolve the 5.6 km/s/Mpc tension. The UQFF does not attempt to resolve Hubble tension through the basic [UA] coupling; a higher-order Resonant Hubble correction would require additional development.

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Source: QCalc_validation.py NED_BASE endpoint | ? = 0.0005/day | [SSq] = 0.57

Appendix: UQFF Production Framework Reference (v4.75+)

> *Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
Îº	5.0 Ã 10â»â´ dayâ»Â¹	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
Î²_i	0.60â0.61	Buoyancy coupling coefficient
kâ	1.5	Ug1 DPM-dipole coupling
kâ	1.2	Ug2 outer-bubble charge coupling
kâ	1.8	Ug3 string-rotation coupling
kâ	2.0	Ug4 vacuum-concentration coupling
Î·	10â»Â²Â²	Inertia tensor scale
E_react(0)	10â´â¶ J	Reference reactive energy

A.2 F_U Master Equation (Complete â 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4} igl[\lambda_i \cdot U_i(r,t) \cdot E_{\text{react}} igr]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
âÎ£Î»áµ¢Â·Uáµ¢Â·E_react	Dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420):
Î»â=10â»Â¹â°, Î»â=10â»Â¹Â², Î»â=10â»Â¹Â¹, Î»â=10â»Â¹Â³ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} imesigl(1 + 10^{13}\,\Theta(ho_{SCm} - ho_c)igr) imesigl(1 + A_q \cos(\Delta\omega\, t)igr)$$

Symbol	Value	Description
Ï_c	10Â¹âµ kg/mÂ³	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
ÎÏ	2Ï/(434Â·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, â¦)	Multi-scale field interactions
Buoyant	Î²_i Ã Ubi	Expanding nebulae, stellar winds
Superconductive	Um Ã (1+10Â¹Â³Â·f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, and `CondensedPhysics2.py`.